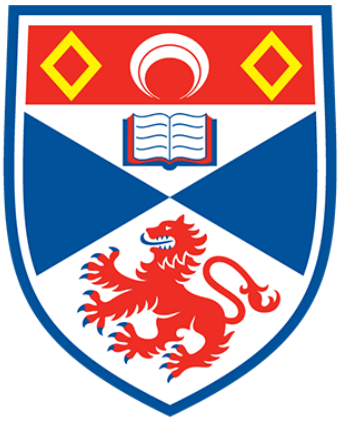
Hausdorff dimension of self-similar sets in complete metric spaces

Barnabas Tanczos



University of
St Andrews

supervised by Jonathan Fraser

October 14, 2026

Abstract

In this paper, we discuss the key characteristics of fractals and introduce the Hausdorff dimension, a well-behaved notion of dimension for fractals based on the Hausdorff measure that permits non-integer values. We develop the theory of iterated function systems as way to encode unique fractal attractors and use these to define self-similar sets. The main theorems provide an elegant formula to compute the Hausdorff dimension of self-similar sets in both arbitrary complete metric spaces and complete doubling spaces.

Declaration

I certify that this project report has been written by me, is a record of work carried out by me, and is essentially different from work undertaken for any other purpose or assessment.

Contents

1 Introduction **1**

2 Hausdorff dimension **3**
2.1 Hausdorff measure 3
2.2 Hausdorff dimension 5
2.3 Properties 6
2.4 Examples 7

3 Iterated function systems **9**
3.1 Contractions 9
3.2 Hausdorff metric 10
3.3 Iterated function systems 14
3.4 Examples 16

4 Self-similar sets **17**
4.1 Mass distribution 18
4.2 Dimension in complete spaces 21
4.3 Dimension in complete doubling spaces 25
4.4 Examples 28

5 Conclusion **32**

References **33**

1 Introduction

Until recently, much of mathematics has been concerned with sets and functions with regular properties that can be analysed using the tools of classical calculus and geometry. Complex objects with non-smooth surfaces were considered pathological and rarely of interest. In 1975, Polish mathematician Benoit Mandelbrot pioneered the study of roughness in his book *Fractals: Form, Chance and Dimension* and coined the term 'fractals'. These fractals reflect a variety of natural phenomena, such as topographical features, plant growth, and cellular structures, and have been found to possess many interesting characteristics. Ever since, the field has significantly expanded, and there are an increasing number of applications in biology, physics, and computer science, among others.

Mandelbrot and others proposed multiple definitions of fractals over time; however, each of these were deemed too restrictive and excluded constructions that clearly should be regarded as fractals. Therefore, instead of a strict definition, fractals are commonly described using the following typical characteristics:

(i) fractals are self-similar, in that they resemble themselves at different scales;

(ii) fractals are formed by an infinite iterative process, resulting in intricate detail with arbitrary precision;

(iii) fractals are too irregular to be described using traditional methods, both locally and globally; and

(iv) fractals have properties that do not reflect their topological dimension.

The archetypal example of a fractal is the middle third Cantor set. It is constructed from the unit interval by repeatedly deleting sections, as can be seen in Figure 1. Let C_0 be the interval $[0, 1]$. The next set C_1 is obtained by removing the middle third to give the two intervals $[0, \frac{1}{3}]$ and $[\frac{2}{3}, 1]$. We continue in this manner, such that each C_k consists of 2^k intervals of length 3^{-k}. The Cantor set itself is the intersection $C = \bigcap_{k=0}^{\infty} C_k$, or the limit of the sequence of sets C_k as $k \to \infty$.

Fractals like the Cantor set defy traditional topological notions of dimension. Although C is an uncountably infinite set containing no isolated points, its length is 0 by any reasonable definition, and, as it is totally disconnected, its topological dimension is also 0.



Figure 1: The first seven iterations in the construction of the middle third Cantor set.

This work introduces a more useful notion of dimension for fractals, the Hausdorff dimension, which allows for non-integer dimensions but requires significant effort to calculate. We focus on the self-similar sets, a specific class of fractals described using iterated function systems, and rigorously prove their existence. The main theorems provide a simple and elegant expression for the Hausdorff dimension of such fractals under specific conditions. The middle third Cantor set is used throughout to illustrate these results.

The primary resource for this essay is *Fractal Geometry* by Kenneth Falconer [1]. The general approach and much of the content, including the main proofs in Section 4, is based on this book. However, this work covers details concerning measure theory that are glossed over, which are expanded upon by van den Eijnden [2], and extends the material beyond $\mathbb{R}^N$ to complete metric spaces, with ideas primarily from Edgar [3], Schief [4], and Balogh and Rohner [5].

2 Hausdorff dimension

To begin, we define and explore the properties of the oldest and most important construction of fractal dimension. By inspecting the behaviour of a measure of the fractal at various scales, the Hausdorff dimension provides a well-behaved notion of smoothness that is defined for any set.

2.1 Hausdorff measure

For the purposes of defining the Hausdorff dimension, we first define the concept of an outer measure, which generalizes the familiar ideas of length, area and volume to arbitrary sets.

Definition 2.1. *Let X be a set and $\mathcal{P}(X)$ its power set. An* ***outer measure*** *on X is a function $\mu : \mathcal{P}(X) \to [0, \infty]$ such that*

(i) $\mu(\varnothing) = 0$*;*

(ii) if $A \subseteq B$ are subsets of X, $\mu(A) \leq \mu(B)$; and

(iii) if $\{A_n\}_n$ is a countable set of subsets of X, then $\mu(\bigcup_n A_n) \leq \sum_n \mu(A_n)$.

Thus, outer measures assign a measure 0 to the empty set, and are both monotone and countably subadditive. This last property distinguishes them from proper measures, which are countably additive. The Hausdorff measure is a measure that can be used in all metric spaces.

Definition 2.2. *Let (X, d) be a metric space and let $S \subseteq X$. Denote the diameter of S by*

$$|S| = \sup \{d(x, y) : x, y \in S\}$$

with $\sup \varnothing = 0$*. For $\delta > 0$ and $s \geq 0$ define*

$$\mathcal{H}^s_\delta(S) = \inf \left\{ \sum_{i=1}^{\infty} |U_i|^s : S \subseteq \bigcup_{i=1}^{\infty} U_i \text{ and } |U_i| < \delta \right\}.$$

Then the s-dimensional ***Hausdorff outer measure*** *is*

$$\mathcal{H}^s(S) = \lim_{\delta \to 0} \mathcal{H}^s_\delta(S).$$

The calculation of the measure involves covering a set with smaller sets of diameter at most δ and minimizing the sum of the sth powers of the diameters. As δ decreases, the range of possible covers is reduced, particularly so for

fractals with fine details. Thus, the infinum $\mathcal{H}^s_\delta(S)$ increases and approaches the limit $\mathcal{H}^s(S)$.

The value s signifies the behaviour of the measure under rescaling. Scaling a set S by λ will scale $\mathcal{H}^s(S)$ by a factor of λ^s. This aligns with our intuitive understanding of rescaling in an s-dimensional space. In this sense, s is the perspective, or dimension, from which we consider the set. We can verify that $\mathcal{H}^s$ is indeed a valid outer measure for any s.

Theorem 2.3. *The Hausdorff outer measure is an outer measure on* (X, d).

Proof. Let $s \geq 0$. The function $\mathcal{H}^s(S)$ clearly maps to non-negative values for any $S \subseteq X$. We verify the required properties of an outer measure.

(i) The empty set may be covered by the empty set, which has diameter 0, so $\mathcal{H}^s(\varnothing) = 0$.

(ii) Let A, B be subsets of X. If $A \subseteq B$, then any cover of B is a cover of A, and so $\mathcal{H}^s(A) \leq \mathcal{H}^s(B)$.

(iii) Let $\{A_n\}_n$ be subsets of X and $\epsilon, \delta > 0$. As $\mathcal{H}^s_\delta(A_n)$ is the infinum over covers $\{B_{n,i}\}_{n,i}$ with $|B_{n,i}| < \delta$, we can find a cover with $\sum_i |B_{n,i}|^s < \mathcal{H}^s_\delta(A_n) + \frac{\epsilon}{2^n}$ for all n. Then $\bigcup_{n,i} B_{n,i}$ is a cover of $\bigcup_n A_n$, and so

$$\mathcal{H}^s_\delta\left(\bigcup_n A_n\right) \leq \sum_{n,i} |B_{n,i}|^s \leq \sum_n \left(\mathcal{H}^s_\delta(A_n) + \frac{\epsilon}{2^n}\right) = \sum_n \mathcal{H}^s_\delta(A_n) + \epsilon.$$

Letting $\epsilon \to 0$ and $\delta \to 0$, it follows that

$$\mathcal{H}^s\left(\bigcup_n A_n\right) \leq \sum_n \mathcal{H}^s(A_n). \qquad \square$$

The Hausdorff outer measure may be restricted to measurable subsets of X, notably including all Borel sets, to form the Hausdorff measure. However, the outer measure has all the necessary properties for our purposes.

As each set in the covers may have a different diameter, the Hausdorff measure is often difficult to estimate via computational methods. Thus, other definitions such as the Minkowski-Bouligand dimension, which covers fractals with sets of equal shape and size, are preferred in these cases.

2.2 Hausdorff dimension

The Hausdorff outer measure has a unique characteristic that leads us to the definition of the Hausdorff dimension. We find that, for any set, the measure is a decreasing function of dimension. Further, it is either infinity or zero at all but a single value of the dimension, where it jumps from one to another.

Theorem 2.4. *Let S be a set and $0 < s < t$. If $\mathcal{H}^s(S) < \infty$, then $\mathcal{H}^t(S) = 0$. If $\mathcal{H}^t(S) > 0$, then $\mathcal{H}^s(S) = \infty$.*

Proof. Let S be covered by $\bigcup_{i=1}^{\infty} U_i$ with $|U_i| < \delta$. Then we have

$$\sum_{i=1}^{\infty} |U_i|^t = \sum_{i=1}^{\infty} |U_i|^{t-s} |U_i|^s \leq \delta^{t-s} \sum_{i=1}^{\infty} |U_i|^s,$$

and taking the infinum on both sides,

$$\mathcal{H}^t_\delta(S) \leq \delta^{t-s} \mathcal{H}^s_\delta(S).$$

Taking $\delta \to 0$, if $\mathcal{H}^s(S) < \infty$, it must be that $\mathcal{H}^t = 0$. The second assertion is the contrapositive. □

Intuitively, when a measure with too low of a dimension is used, such length for a plane, the set is infinitely large. Meanwhile, when a measure of too high of a dimension is used, such as volume for a plane, the set is so small it must have measure zero. The boundary point between these is what we define as the Hausdorff dimension of the set.

Definition 2.5. *The **Hausdorff dimension** of a set S is given by*

$$\dim_H S = \inf \{s \geq 0 : \mathcal{H}^s(S) = 0\} = \sup \{s \geq 0 : \mathcal{H}^s(S) = \infty\}.$$

*The set S has **strict Hausdorff dimension** s if*

$$0 < \mathcal{H}^s(S) < \infty.$$

The value of the Hausdorff measure at the dimension of a set may be zero or infinity, but for strict dimension it must be a positive number. In the this case, the finite, positive size of the set relative to its dimension makes it more well-behaved.

2.3 Properties

The Hausdorff dimension has several useful basic properties, which we ideally expect for any reasonable definition of dimension in order to align with intuition and other traditional dimensions. They provide a base case where countable sets always have dimension 0, and confirm that the dimension is both monotonic and countably stable. The properties reflect those of the underlying outer measure.

Theorem 2.6.

(i) Let A, B be subsets of X. If $A \subseteq B$, then $\dim_H A \leq \dim_H B$.

(ii) Let $\{A_n\}$ be subsets of X. Then $\dim_H \left(\bigcup_n A_n\right) = \sup_n\{\dim_H A_n\}$.

(iii) Let A be a countable subset of X. Then $\dim_H A = 0$.

Proof.

(i) This follows from the measure property that $\mathcal{H}^s(A) \leq \mathcal{H}^s(B)$.

(ii) From property (i), it is clear that $\dim_H \left(\bigcup_n A_n\right) \geq \dim_H A_n$ for all n, and so $\dim_H \left(\bigcup_n A_n\right) \geq \sup_n\{\dim_H A_n\}$. For the opposite inequality, if $s > \dim_H A_n$ for all n, then $\mathcal{H}^s(A_n) = 0$ and $\mathcal{H}^s\left(\bigcup_n A_n\right) \leq \sum_n \mathcal{H}^s(A_n) = 0$. Therefore, we have $\dim_H \left(\bigcup_n A_n\right) \leq \sup_n\{\dim_H A_n\}$.

(iii) Let $A = \bigcup_n A_n$, where each A_n contains a single point. We can cover each A_n with a set of diameter 0, so $\mathcal{H}^0(A_n) = 1$ and $\dim_H A_n = 0$. By property (ii), $\dim_H A = \dim_H \left(\bigcup_n A_n\right) = \sup_n\{A_n\} = 0$. □

An upper bound for the dimension of a set can usually be found in a straightforward way by finding an effective covering with small sets. The lower bound can be significantly more difficult to estimate, as it involves showing that there does not exist any covering that provides a lower value for the measure. A useful tool for this is the following theorem. It involves a type of measure called a mass distribution, which divides a finite mass over the measured sets.

Definition 2.7. *A **mass distribution** on a set S is an outer measure μ that satisfies $0 < \mu(S) < \infty$.*

Using this construct, we can ensure that the mass $\mu(U)$ of a covering set U, and so its contribution to covering S, cannot be too large compared to its size measured as $|U|^s$. Only a set that is large enough can hold the mass without concentrating too much in a particular region.

Theorem 2.8. *Let μ be a mass distribution on S and $s > 0$. Suppose there is $\epsilon, c > 0$ such that*

$$\mu(U) \leq c|U|^s$$

for any set U with $|U| < \epsilon$. Then

$$\mathcal{H}^s(S) \geq \frac{\mu(S)}{c} > 0$$

and thus

$$s \leq \dim_H S.$$

Proof. Let $\epsilon > \delta > 0$ and let $\bigcup_{i=1}^{\infty} U_i$ be a cover of S with $|U_i| < \delta < \epsilon$. Then $\mu(U_i) \leq c|U_i|^s$, and so

$$0 < \mu(S) \leq \mu\left(\bigcup_{i=1}^{\infty} U_i\right) \leq \sum_{i=1}^{\infty} \mu(U_i) \leq c \sum_{i=1}^{\infty} |U_i|^s \leq c\mathcal{H}^s_\delta(S)$$

using the properties of outer measures. Thus

$$\mathcal{H}^s_\delta(S) \geq \frac{\mu(S)}{c}$$

for all $\epsilon > \delta > 0$ and the result follows by taking $\delta \to 0$. □

This result will be crucial in proving a lower bound for the results in Section 4. An important part of the proofs is defining a useful mass distribution over a given fractal.

2.4 Examples

The middle third Cantor set illustrates the comparative difficulty in finding a lower bound for the Hausdorff measure. We will show that the Hausdorff dimension of the set C is $s = \log 2/\log 3$. Recall that $C = \bigcap_{k=0}^{\infty} C_k$, where each C_k consists of 2^k intervals of length 3^{-k}.

For the upper bound, let $\delta > 0$. We cover the set by the natural cover C_k, where $k \in \mathbb{N}$ is chosen such that the diameter of each interval $3^{-k} < \delta$. This implies that $\mathcal{H}^s_\delta(C) \leq 2^k 3^{-ks} = 2^k 2^{-k} = 1$ using the above value of s. Taking $\delta \to 0$ gives $\mathcal{H}^s(C) \leq 1$.

For the lower bound, we use the mass distribution principle to simplify the calculation, which would otherwise be quite tedious even for such a simple fractal. Let μ be the natural mass distribution on C, where we divide unit

mass on C_0 evenly between the two intervals in the next layer. Thus, each of the 2^k intervals in C_k have mass 2^{-k}. This can straightforwardly be formalised as a valid mass distribution. Let U be a set with $|U| < 1$ and choose $k \in \mathbb{N}$ such that $3^{-(k+1)} \leq |U| < 3^{-k}$. Then U can intersect at most one interval in C_k, so

$$\mu(U) \leq 2^{-k} = 3^{-ks} = 3^s 3^{-(k+1)s} \leq 3^s |U|^s,$$

and thus $\mathcal{H}^s(C) \geq 3^{-s} = \frac{1}{2}$.

As the s-dimensional measure is both non-zero and finite, we conclude that indeed $\dim_H C = \log 2 / \log 3$, and that this dimension is strict. Notice that this result is related to the fact that at each step of the construction, an interval is divided into two intervals each scaled by a third. This observation is made more rigorous in Section 4.

3 Iterated function systems

An important property of many fractals is that they resemble themselves at different scales. This can be represented by contracting functions, which are iteratively applied to produce increasingly finer details until they reach a fixed point. Iterated function systems are families of such contractions. They can represent many fractals in a straightforward manner, and may be used to deduce important properties.

3.1 Contractions

The key components of iterated function systems are contractions, which are functions that reduce the distance between points in the space.

Definition 3.1. *Let (X, d) be a metric space. A function $f : X \to X$ is a* ***contraction*** *on X if there is a ratio $0 \leq c < 1$ such that*

$$d(f(x), f(y)) \leq cd(x, y)$$

for all $x, y \in X$.

As the distance between any two points is scaled by c, we have that for any set U the diameter $|f(U)| \leq c|U|$. Additionally, the composition of contractions $f_1, \ldots, f_n$ with ratios $c_1, \ldots, c_n$ is also a contraction with ratio $c_1 \cdots c_n$.

Therefore, we can apply contractions in a iterative fashion, denoting the nth iterate of f with f^n, where $f^0(x) = x$ and $f^n(x) = f(f^{n-1}(x))$ for $n \geq 1$. As the function continually decreases the distance between points, we eventually find a point where the distance is 0, so that the point is mapped to itself. In a complete metric space, such a fixed point is always present and unique.

Theorem 3.2. *Let f be a contraction on a non-empty complete metric space (X, d). Then f has a unique fixed point $x_0 \in X$ where $f(x_0) = x_0$. Furthermore, $d(f^n(x), x_0) \to 0$ as $n \to \infty$ for all $x \in X$.*

Proof. Let $x \in X$ and c be the ratio of f. We will show that $(f^n(x))_n$ converges and that its limit is a fixed point. As the metric space (X, d) is complete, it suffices to show that the sequence is Cauchy to prove convergence.

For all $n \in \mathbb{N}$, we can apply the definition of a contraction to get

$$d(f^n(x), f^{n-1}(x)) \leq cd(f^{n-1}(x), f^{n-2}(x)) \leq \ldots \leq c^{n-1}d(f(x), x).$$

Now let $n, m \in \mathbb{N}$ with $n \geq m$. Using the triangle inequality repeatedly,

$$d(f^n(x), f^m(x)) \leq d(f^n(x), f^{n-1}(x)) + d(f^{n-1}(x), f^{n-2}(x)) + \ldots + d(f^{m+1}(x), f^m(x)$$

By applying the first inequality to each term and summing the geometric series, we have

$$\begin{aligned} d(f^n(x), f^m(x)) &\leq (c^{n-1} + c^{n-2} + \ldots + c^m) d(f(x), x) \\ &\leq \frac{c^m}{1-c} d(f(x), x) \\ &\to 0 \end{aligned}$$

as $n, m \to \infty$. Thus $(f^n(x))_n$ is Cauchy and converges to some $x_0 \in X$. As f is continuous, we then have

$$f^{n+1}(x) \to x_0$$

$$f^{n+1}(x) = f(f^n(x)) \to f(x_0)$$

as $n \to \infty$. By uniqueness of limits, $f(x_0) = x_0$. Thus, x_0 is a fixed point of f. It is unique, because if x_1 is another fixed point, then

$$d(x_0, x_1) = d(f(x_0), f(x_1)) \leq c d(x_0, x_1),$$

which is possible only if $d(x_0, x_1) = 0$ and thus $x_0 = x_1$. To show the last statement, we have that for any $x \in X$,

$$d(f^n(x), x_0) = d(f^n(x), f^n(x_0)) \leq c^n d(x, x_0) \to 0$$

as $n \to \infty$. □

The last property is frequently useful for iteratively finding the fixed point guaranteed by the theorem, especially because the starting point may be any point in the space.

3.2 Hausdorff metric

To extend Theorem 3.2 to find fixed points that are fractal sets, we first introduce the Hausdorff metric, which provides a way to measure the distance between non-empty compact sets in an arbitrary metric space.

Definition 3.3. *The* ϵ***-neighbourhood*** *of a set* $A \subseteq X$ *is defined as*

$$A + \epsilon = \{x \in X : \exists a \in A \text{ with } d(x, a) < \epsilon\}.$$

Definition 3.4. *Let (X, d) be a metric space and $H(X)$ be the set of non-empty compact subsets of X. The* ***Hausdorff distance*** *between sets $A, B \in H(X)$ is defined as*

$$D(A, B) = \inf \{\epsilon > 0 : A \subseteq B + \epsilon \text{ and } B \subseteq A + \epsilon\}.$$

In other words, the Hausdorff distance is the distance by which both sets must be enlarged around their edges such that they contain each other. Alternatively, it can be construed as the greatest distance between a point in one set and the closest point in the other set. The Hausdorff distance has a useful property that allows us to, for any point in a set, find a point in the other set that is sufficiently close.

Lemma 3.5. *Let $A, B \in H(X)$. For all $a \in A$, there is $b \in B$ such that $d(a, b) \leq D(A, B)$.*

Proof. For any $\epsilon > 0$, we have $A \subseteq B + D(A, B) + \epsilon$, which implies that $d(a, b_\epsilon) < D(A, B) + \epsilon$ for some $b_\epsilon \in B$. Taking $\epsilon \to 0$, the sequence of b_ϵ have a limit point b satisfying $d(a, b) \leq D(A, B)$. Since B is compact and thus closed, $b \in B$. □

The Hausdorff distance induces a metric space on the non-empty compact subsets of any metric space. These criteria for the sets are appropriate, as the empty set has no neighbourhood and so the distance to any other set would be undefined or infinite. Compactness, meanwhile, implies closed and bounded in particular, and is necessary to guarantee the properties of the metric space.

Theorem 3.6. *The Hausdorff distance is a metric on $H(X)$.*

Proof. Let $A, B, C \in H(X)$. We must show that D satisfies the standard requirements.

(i) By definition, it is clear that $D(A, B) \geq 0$. As A and B are compact and thus bounded, all of their points are within a finite distance and $D(A, B) < \infty$.

(ii) If $A = B$, then for every $\epsilon > 0$ it holds that $A \subseteq B + \epsilon$ and $B \subseteq A + \epsilon$, implying $D(A, B) = 0$. Conversely, if $D(A, B) = 0$, consider any $a \in A$. From the property above, $d(a, b) \leq D(A, B) = 0$ for some $a = b \in B$. It follows that $A \subseteq B$, and similarly $B \subseteq A$. We conclude that $A = B$.

(iii) By the symmetrical construction $D(A, B) = D(B, A)$.

(iv) Let $\epsilon > 0$. For $a \in A$, there is $b \in B$ such that $d(a,b) \leq D(A,B)$. Then, there is $c \in C$ with $d(b,c) \leq D(B,C)$. Since

$$d(a,c) \leq d(a,b) + d(b,c) < D(A,B) + D(B,C) + \epsilon,$$

we have the neighbourhood

$$A \subseteq C + D(A,B) + D(B,C) + \epsilon,$$

and similarly

$$C \subseteq A + D(A,B) + D(B,C) + \epsilon.$$

Therefore,

$$D(A,C) \leq D(A,B) + D(B,C) + \epsilon,$$

and taking $\epsilon \to 0$, we have the triangle inequality

$$D(A,C) \leq D(A,B) + D(B,C). \qquad \square$$

In order to apply Theorem 3.2, we require a complete metric space. Therefore, we prove the following additional feature of the induced metric space. If a sequence of sets is Cauchy with respect to the Hausdorff metric, its points will also form Cauchy sequences, and so the sets will converge to the set of limit points of those sequences. Thus, the completeness property is transferred to the induced space.

Theorem 3.7. *If X is a complete metric space, then the space $(H(X), D)$ is also complete.*

Proof. Let $(A_n)_n$ be a Cauchy sequence in $H(X)$. We will prove it converges to the set

$$A = \{x \in X : \exists (x_n)_n \text{ with } x_n \in A_n \text{ and } x_n \to x\},$$

that is, for all $\epsilon > 0$, there is $n \in \mathbb{N}$ such that $A \subseteq A_n + \epsilon$ and $A_n \subseteq A + \epsilon$. Then, we will show that A is non-empty and compact, and so an element of $H(X)$. A set is compact if and only if it is complete and totally bounded, and a subset of a complete set is complete if and only if it is closed. Thus, it suffices to show that A is non-empty, closed, and totally bounded.

(i) Let $\epsilon > 0$. As $(A_n)_n$ is Cauchy, there is $N \in \mathbb{N}$ such that $n, m \geq N$ implies that $D(A_n, A_m) < \frac{\epsilon}{2}$. Let $n \geq N$. From the definition of the Hausdorff distance, $D(A_n, A) \leq \epsilon$ only if $A \subseteq A_n + \epsilon$ and $A_n \subseteq A + \epsilon$.

Let $x \in A$. From the construction of A, there is a sequence $(x_k)_k$ with $x_k \in A_k$ and $x_k \to x$. Thus, there is a $K \in \mathbb{N}$ such that $d(x_k, x) < \frac{\epsilon}{2}$ for all $k \geq K$. Let $k \geq N, K$. Then $D(A_k, A_n) < \frac{\epsilon}{2}$ implies that there is a $y \in A_n$ with $d(x_k, y) < \frac{\epsilon}{2}$. Therefore,

$$d(x, y) \leq d(x, x_k) + d(x_k, y) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon,$$

which shows that $A \subseteq A_n + \epsilon$.

Now let $y \in A_n$. We can choose $n = K_1 < K_2 < \ldots$ such that $D(A_{K_i}, A_m) < \frac{\epsilon}{2^i}$ for all $m \geq K_i$. This can be used to define a sequence $(y_k)_k$ with $y_k \in A_k$. First, for $k < n$, choose $y_k \in A_k$ arbitrarily and set $y_{K_1} = y$. Then, for a fixed y_{K_i}, we can choose all $K_i < k \leq K_{i+1}$ such that $y_k \in A_k$ and $d(y_{K_i}, y_k) \leq D(A_{K_i}, A_k) < \frac{\epsilon}{2^i}$.

This sequence is Cauchy. Letting $\delta > 0$, take $D(A_{K_i}, A_m) < \frac{\delta}{3}$ for all $m \geq K_i$ in the above. For any $l, m > n$, we have $K_i < l \leq K_{i+1}$ and $K_j < m < K_{j+1}$, so

$$d(y_l, y_m) \leq d(y_l, y_{K_i}) + d(y_{K_i}, y_{K_j}) + d(y_{K_j}, y_m) < \delta.$$

Hence, $(y_k)_k$ converges to some $x \in A$. For any $k > n$, we have $K_i < k \leq K_{i+1}$, giving

$$\begin{aligned} d(y, y_k) &\leq d(y, y_{K_1}) + d(y_{K_1}, y_{K_2}) + \ldots + d(y_{K_i}, y_k) \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2^2} + \ldots + \frac{\epsilon}{2^i} < \epsilon. \end{aligned}$$

Since $d(y, y_k) \to d(y, x)$, it must be that $d(y, x) \leq \epsilon$ and so $A_n \subseteq A + \epsilon$.

(ii) Taking any ϵ in the above argument, and using that each A_k is non-empty, we can construct a convergent sequence $(y_k)_k$ with $y_k \in A_k$. By the definition of A, it contains the limit of that sequence and thus is non-empty.

(iii) Let x be a limit point of A. Then there exists a sequence $(x_n)_n$ in A with $d(x_n, x) \to 0$. Also, as $A_n \to A$, there is a point $y_n \in A_n$ such that $d(x_n, y_n) \leq D(A, A_n) \to 0$. Hence,

$$d(y_n, x) \leq d(y_n, x_n) + d(x_n, x) \to 0,$$

and so $y_n \to x$ as $n \to \infty$. This shows from the definition of A that $x \in A$. Therefore, A must be closed.

(iv) Let $\epsilon > 0$. As $A_n \to A$, we can choose $n \in \mathbb{N}$ such that $D(A_n, A) < \frac{\epsilon}{3}$. The set A_n is compact and thus totally bounded, so there exists a finite cover $\bigcup_{i=1}^{m} B(y_i, \epsilon)$ with $y_i \in A_n$. For each y_i, there is $x_i \in A$ with $d(x_i, y_i) < \frac{\epsilon}{3}$. We show that $\bigcup_{i=1}^{m} B(x_i, \epsilon)$ is a finite cover of A. Let $x \in A$. There exists a $y \in A_n$ such that $d(x, y) \leq D(A, A_n) < \frac{\epsilon}{3}$. Thus, for some $1 \leq i \leq m$, we have

$$d(x, x_i) \leq d(x, y) + d(y, y_i) + d(y_i, x_i) < \epsilon.$$

We conclude that any Cauchy sequence $(A_n)_n$ indeed converges to a set in $H(X)$, and so $(H(X), D)$ is complete. □

3.3 Iterated function systems

A single contraction has a fixed point that is a single element. In order to represent fractal sets, we require a collection of contractions called an iterated function system. Similarly to the singular case, these give rise to a unique set that remains invariant under the system. Even a small set of contractions can represent many complex fractals.

Definition 3.8. *An **iterated function system (IFS)** is a finite family of contractions $\{f_1, \ldots, f_n\}$ with $n > 1$.*

Definition 3.9. *A set S is called the **attractor** for an IFS $\{f_1, \ldots, f_n\}$ if*

$$S = f(S) = \bigcup_{i=1}^{n} f_i(S).$$

Using the Hausdorff metric to measure the distances between sets, we will now use Theorem 3.2 to show that any IFS also has a unique fixed point, namely the set that is its attractor. This allows us to completely specify a fractal using only the corresponding IFS.

Theorem 3.10. *Let (X, d) be a non-empty complete metric space. An iterated function system $\{f_1, \ldots, f_n\}$ on X has a unique non-empty compact attractor.*

Proof. Consider the induced space $H(X)$ with the Hausdorff metric D. Since X is complete, so is $H(X)$, by Theorem 3.7. Define $f : H(X) \to H(X)$ by

$$f(A) = \bigcup_{i=1}^{n} f_i(A).$$

This function does indeed map compact sets onto other compact sets. The contractions f_i are clearly continuous, and the continuous image of a compact set, as well as the union of finitely many compact sets, is compact.

Notice that the fixed point of f is exactly the attractor of the IFS. In order to apply Theorem 3.2, we show that f is also a contraction.

Let $0 \leq c_i < 1$ be the ratios of the contractions f_i and set $c = \max_i c_i$, so that $0 \leq c < 1$. Let $A, B \in H(X)$. Each point in the image $a' \in f(A)$ has a corresponding $a \in A$ such that $a' = f(a) = f_i(a)$ for at least some i. The Hausdorff distance ensures there is always a sufficiently close point in the other set: by Lemma 3.5, there is $b \in B$ that satisfies $d(a, b) \leq D(A, B)$. For this b, set $b' = f_i(b) \in f(B)$. Then we have

$$d(a', b') = d(f_i(a), f_i(b)) \leq c_i d(a, b) < cD(A, B) + \epsilon.$$

As $a' \in f(A)$ was arbitrary,

$$f(A) \subseteq f(B) + cD(A, B) + \epsilon$$

and similarly

$$f(B) \subseteq f(A) + cD(A, B) + \epsilon.$$

Thus, the Hausdorff distance

$$D(f(A), f(B)) \leq cD(A, B) + \epsilon,$$

and taking $\epsilon \to 0$,

$$D(f(A), f(B) \leq cD(A, B).$$

Therefore, f is a contraction on the complete metric space $H(X)$. By Theorem 3.2, f has a unique fixed point S with $f(S) = S$ that is the non-empty compact attractor for the IFS. □

Finding the right contractions is usually straightforward for fractals with a self-similar pattern. Then, we can leverage the second part of Theorem 3.2 to decode the attractor S and its elements as an intersection of contractions applied to an arbitrary set. We use the notation $U_{i_1,\ldots,i_k} = f_{i_1} \circ \ldots \circ f_{i_k}(U)$.

Corollary 3.11. *Let $U \in H(X)$ with $f(U) \subseteq U$. Then*

$$S = \bigcap_{k=1}^{\infty} f^k(U)$$

and any $x \in S$ can be encoded as $x_{i_1,i_2,\ldots}$ where

$$\{x\} = \bigcap_{k=1}^{\infty} U_{i_1,\ldots,i_k}.$$

Proof. For the first part, we know from Theorem 3.2 that $f^k(U) \to S$ as $k \to \infty$. A convergent sequence of non-empty compact sets necessarily has a non-empty compact intersection. Moreover, if $f(U) \subseteq U$, then $f^k(U) \subseteq f^{k-1}(U)$ for all k, so $f^k(U)$ is a decreasing sequence. This implies $S \subseteq f^k(U)$ for all k, and as such, that S is that intersection.

For the second part, $U_{i_1,\ldots,i_k}$ is also a decreasing sequence of compact sets, and as $|U_{i_1,\ldots,i_k}| \to 0$ as $k \to \infty$, the non-empty intersection must be a single point $x_{i_1,\ldots,i_k}$. This point is clearly in S, as

$$\bigcap_{k=1}^{\infty} U_{i_1,\ldots,i_k} \subseteq \bigcap_{k=1}^{\infty} f^k(U) = S.$$

Finally, every point in S can be represented in this way, as for all k,

$$f^k(U) = \bigcup_{i_1,\ldots,i_k} U_{i_1,\ldots i_k}. \qquad \square$$

Note that this encoding is not necessarily unique, but linking any point in S with a corresponding sequence of contractions can be highly useful.

3.4 Examples

The middle third Cantor set can easily be expressed as an IFS on the unit interval. We define the contractions $f_1, f_2 : [0,1] \to [0,1]$ by

$$f_1(x) = \frac{1}{3}x \quad \text{and} \quad f_2(x) = \frac{1}{3}x + \frac{2}{3}.$$

The two functions f_1 and f_2 represent the left and right parts of the fractal C, respectively. Any closed interval $[a,b]$ in the domain is a compact subset, so C can be obtained by

$$C = \bigcap_{k=1}^{\infty} f^k([a,b]).$$

Notice that these specific contractions scale every point in the set equally by $\frac{1}{3}$. In the following section, we will focus on these well-behaved functions.

4 Self-similar sets

In this section, we focus on a specific set of iterated function systems, namely those where the contractions scale by a constant factor.

Definition 4.1. *Let (X, d) be a metric space. A function $f : X \to X$ is a* ***similarity*** *on X if there is a ratio $0 \leq c < 1$ such that*

$$d(f(x), f(y)) = cd(x, y)$$

for all $x, y \in X$.

As similarities are contractions, they can form an IFS and all the previous theorems apply to these. Thus we have the following special attractor.

Definition 4.2. *A* ***self-similar set*** *is the attractor of an IFS consisting of similarities.*

Self-similar sets such as the Cantor set are well-behaved precisely due to their highly uniform self-similarity. For a self-similar fractal S, if the results of the initial mappings $S_1, \ldots, S_n$ are 'nearly' disjoint, we can estimate the Hausdorff dimension with the heuristic calculation

$$\mathcal{H}^s(S) = \mathcal{H}^s\left(\bigcup_{i=1}^{n} f_i(S)\right) = \sum_{i=1}^{n} \mathcal{H}^s(f_i(S)) = \sum_{i=1}^{n} c_i^s \mathcal{H}^s(S),$$

assuming that the Hausdorff measure is additive for these sets. In the case that $0 < \mathcal{H}^s(S) < \infty$, we can divide either side to get $\dim_H S = s$, where s satisfies $\sum_{i=1}^{n} c_i^s = 1$. This prompts the following definition.

Definition 4.3. *Let $\{f_1, \ldots, f_n\}$ be an IFS that has ratios $c_1, \ldots, c_n$. The* ***similarity value*** *s is given by*

$$\sum_{i=1}^{n} c_i^s = 1.$$

We will show that in many common cases, the Hausdorff dimension is indeed equal to or bounded by the similarity value, and so can elegantly be expressed by this easily calculable formula.

4.1 Mass distribution

The results in this section rely on estimating the lower bound for the Hausdorff measure of an arbitrary self-similar fractal using Theorem 2.8. This requires a suitable mass distribution, which we will construct rigorously here.

The set of concern is the infinite product space $X^{\mathbb{N}} = X \times X \times \cdots$ for a particular set X, that is, the set of infinite sequences where each term is in X. We denote the finite product space in the same way as $X^k = X \times \cdots \times X$, where the Cartesian product is taken $k \in \mathbb{N}$ times. These will represent the similarity ratios.

First, we define a pre-measure on a well-behaved family of subsets, then extend it to form the necessary outer measure.

Definition 4.4. *A family of sets $\mathcal{R}$ is a **ring** if for all $A, B \in \mathcal{R}$,*

(i) $A \cup B \in \mathcal{R}$*; and*

(ii) $A \setminus B \in \mathcal{R}$*.*

A ring of subsets of $X^{\mathbb{N}}$ consists of sets of 'cylinders', which contain all infinite sequences starting with a specific finite sequence.

Proposition 4.5. *Let X be a finite set. For $k \in \mathbb{N}$ and $A \subseteq X^k$ define*

$$C_A = \big\{(x_1, \ldots, x_k, \ldots) \in X^{\mathbb{N}} : (x_1, \ldots, x_k) \in A\big\}$$

with $C_\varnothing = X^{\mathbb{N}}$. Then the family

$$\mathcal{C} = \big\{C_A : k \in \mathbb{N}, A \subseteq X^k\big\}$$

is a ring.

Proof. We show that $\mathcal{C}$ satisfies the two requirements:

(i) Let $A \subseteq X^k$ and $B \subseteq X^l$. If $k = l$, then $A \cup B \subseteq X_k$ and so $C_A \cup C_B = C_{A \cup B} \in \mathcal{C}$. If $l > k$, then C_A is equivalent to $C_{A'}$, where $A' = A \times X^{l-k}$. As $A' \subseteq X^l$, we can show using the first case that $C_A \cup C_B \in \mathcal{C}$ also.

(ii) This follows in a similar manner, as $C_A \setminus C_B = C_{A \setminus B}$ if $A, B \subseteq X^k$. □

We create a pre-measure on this ring, which is a less strict precursor to a proper (outer) measure on the entire set.

Definition 4.6. *Let $\mathcal{R}$ be a ring. A* ***pre-measure*** *on $\mathcal{R}$ is a function $\mu : \mathcal{R} \to [0, \infty]$ such that*

(i) $\mu(\varnothing) = 0$; and

(ii) if $\{A_n\}_n$ is a countable family of pairwise disjoint subsets of $\mathcal{R}$ with $\bigcup_n A_n \in \mathcal{R}$, then $\mu(\bigcup_n A_n) = \sum_n \mu(A_n)$.

We construct a pre-measure that assigns each element in X a weight such that all weights add up to 1. The measure of a cylinder is the product of these weights, and the measure of a set of cylinders is the sum of their individual measures.

Proposition 4.7. *Let $p : X \to [0, 1]$ be a function defined by $p(x) = p_x$ such that $\sum_{x \in X} p_x = 1$. For $k \in \mathbb{N}$ and $A \subseteq X^k$ define*

$$\mu(C_A) = \sum_{(x_1, \ldots, x_k) \in A} p_{x_1} \cdots p_{x_k}.$$

Then μ is a pre-measure on $\mathcal{C}$ with $\mu(X^{\mathbb{N}}) = 1$.

Proof. First, we verify the two criteria of a premeasure:

(i) It is clear that $\mu(\varnothing) = 0$.

(ii) Let $\{C_{A_n}\}_n$ be a countable family of pairwise disjoint subsets of $\mathcal{C}$ with $\bigcup_n C_{A_n} = C_{\cup_n A_n} \in \mathcal{C}$. This implies $\bigcup_n A_n \subseteq X^k$ for some $k \in \mathbb{N}$. Therefore, each $A_n \subseteq X_k$ and

$$\begin{aligned} \mu\left(\bigcup_n C_{A_n}\right) &= \mu(C_{\cup_n A_n}) \\ &= \sum_{\cup_n A_n} p_{x_1} \cdots p_{x_k} \\ &= \sum_n \sum_{A_n} p_{x_1} \cdots p_{x_k} \\ &= \sum_n \mu(C_{A_n}). \end{aligned}$$

Separating the summation is possible because the A_n are also pairwise disjoint.

It can then be seen that

$$\mu(X^{\mathbb{N}}) = \mu\left(\bigcup_{x \in X} C_{\{(x)\}}\right) = \sum_{x \in X} \mu(C_{\{(x)\}}) = \sum_{x \in X} p_x = 1. \qquad \square$$

This can be extended to an outer measure on the entirety of $X^{\mathbb{N}}$ by finding the infinum of covers using sets of cylinders for any arbitrary set. The result follows directly from Carathéodory's extension theorem, but we prove it here for this specific case.

Proposition 4.8. *Let $A \subseteq X^{\mathbb{N}}$. Then the function μ^* defined by*

$$\mu^*(A) = \inf\left\{\sum_{i=1}^{\infty} \mu(B_i) : B_i \in \mathcal{C}, A \subseteq \bigcup_{i=1}^{\infty} B_i\right\}$$

is an outer measure on $X^{\mathbb{N}}$.

Proof. We show the required properties of an outer measure:

(i) The empty set $\varnothing$ is covered by $\varnothing$, so $\mu^*(\varnothing) \leq \mu(\varnothing) = 0$.

(ii) If $A \subseteq B$ in $X^{\mathbb{N}}$, then any cover of B is also a cover of A, directly implying $\mu^*(A) \leq \mu^*(B)$.

(iii) Let $\{A_n\}_n$ be a countable family of subsets of $X^{\mathbb{N}}$ and $\epsilon > 0$. By definition, we can find a cover $\{B_{n,i}\}_i$ in $\mathcal{C}$ for each A_n such that $\sum_i \mu(B_{n,i}) < \mu^*(A_n) + \frac{\epsilon}{2^n}$. Then $\{B_{n,i}\}_{n,i}$ forms a countable cover of $\bigcup_n A_n$ and so

$$\mu^*\left(\bigcup_n A_n\right) \leq \sum_{n,i} \mu(B_{n,i}) < \sum_n \left(\mu^*(A_n) + \frac{\epsilon}{2^n}\right) = \epsilon + \sum_n \mu^*(A_n).$$

The property follows by taking $\epsilon \to 0$. □

Lastly, we show that the two measures coincide on $\mathcal{C}$, so we can easily find the outer measure of a cylinder set.

Proposition 4.9. *For any $A \in \mathcal{C}$, $\mu^*(A) = \mu(A)$.*

Proof. Let $A \in \mathcal{R}$. Taking A as the cover of A, it is immediate that $\mu^*(A) \leq \mu(A)$. For the reverse inequality, we use the monotonicity and countable subadditivity of pre-measures, which may quickly be derived from its definition:

(i) if $A \subseteq B$, then $\mu(A) \leq \mu(B)$; and

(ii) if $\{A_n\}_n$ is a countable family of subsets of $\mathcal{R}$ with $\bigcup_n A_n \in \mathcal{R}$, then $\mu(\bigcup_n A_n) \leq \sum_n \mu(A_n)$.

Consider an arbitrary cover $\{B_i\}_i$ of A. Then $A = A \cap (\bigcup_i B_i) = \bigcup_i (A \cap B_i)$, and so

$$\mu(A) = \mu\left(\bigcup_i (A \cap B_i)\right) \leq \sum_i \mu(A \cap B_i) \leq \sum_i \mu(B_i).$$

As the chosen cover was arbitrary, it must be that $\mu^*(A) \geq \mu(A)$. □

This gives us the mass distribution that we require, which will enable us to divide a finite mass between the infinite sequences of similarities that constitute a fractal.

Corollary 4.10. *The outer measure μ^* forms a mass distribution on $X^{\mathbb{N}}$.*

Proof. This is a consequence of Proposition 4.9, as $\mu^*(X^{\mathbb{N}}) = \mu(X^{\mathbb{N}}) = 1$. □

4.2 Dimension in complete spaces

As before, it is relatively easy to show that the similarity value provides an upper bound for the Hausdorff dimension of a self-similar set. In fact, we can provide a generalised result that applies to any IFS. We denote the index set $\{1, \ldots, n\}$ for an IFS by I.

Theorem 4.11. *Let $\{f_1, \ldots, f_n\}$ be an IFS with attractor S and let s be given by*

$$\sum_{i=1}^{n} c_i^s = 1,$$

where $c_1, \ldots, c_n$ are the contraction ratios of the IFS. Then $\dim_H S \leq s$.

Proof. Let $\delta > 0$. Consider the cover

$$S \subseteq \bigcup_{I^k} S_{i_1,\ldots,i_k}$$

given by repeated applications of the definition of an attractor. Because $f_{i_1} \circ \ldots \circ f_{i_k}$ is a similarity of ratio $c_{i_1} \cdots c_{i_k}$, we can choose k such that $|S_{i_1,\ldots,i_k}| \leq c_{i_1} \cdots c_{i_k} |S| \leq (\max_i c_i)^k |S| < \delta$.

Then we have

$$\begin{aligned}\mathcal{H}^s_\delta(S) &\leq \sum_{I^k} |S_{i_1,\ldots,i_k}|^s \\ &\leq \sum_{I^k} (c_{i_1}\cdots c_{i_k})^s |S|^s \\ &= \left(\sum_I c_{i_1}^s\right)\cdots\left(\sum_I c_{i_k}^s\right)|S|^s \\ &= |S|^s\end{aligned}$$

using the definition of s. Taking $\delta \to 0$, we have shown the upper bound $\mathcal{H}^s(S) \leq |S|^s < \infty$. Therefore, $\dim_H S \leq s$. □

Proving a lower bound is less simple. We first show this with the very strong condition that the contractions map onto pairwise disjoint sets. The result holds for any IFS consisting of contractions that are limited in how fast they reduce the distance between points, which includes similarities.

Theorem 4.12. *Let $\{f_1, \ldots, f_n\}$ be an IFS with attractor S and*

$$b_i d(x, y) \leq d(f_i(x), f_i(y))$$

for all $x, y \in X$ and some $0 \leq b_i < 1$. Let s be given by

$$\sum_{i=1}^n b_i^s = 1.$$

If $S_1, \ldots, S_n$ are pairwise disjoint, then $\dim_H S \geq s$.

Proof. Let $d > 0$ be the minimum distance between disjoint sets $S_1, \ldots, S_n$, that is

$$d = \min_{i \neq j} \inf \{d(x, y) : x \in S_i, y \in S_j\}.$$

For $A \subseteq X$, define the function ν by

$$\nu(A) = \mu^*(\{(i_1, i_2, \ldots) : x_{i_1,i_2,\ldots} \in A \cap S\}),$$

where μ^* is constructed as in Proposition 4.8, with $X = I$ and $p(i) = b_i^s$. This is an outer measure, and forms a mass distribution as $\nu(S) = \mu^*(X^{\mathbb{N}}) = 1$. Also note that $\nu(S_{i_1,\ldots,i_k}) = \mu^*(C_{\{(i_1,\ldots,i_k)\}}) = \mu(C_{\{(i_1,\ldots,i_k)\}}) = (b_{i_1}\cdots b_{i_k})^s$. We will use this to apply Theorem 2.8.

Let $x \in S$ and $0 < r < d$. Then $x = x_{i_1,i_2,\ldots}$ for some $(i_1, i_2, \ldots) \in I^{\mathbb{N}}$, and $x \in S_{i_1,\ldots,i_k}$ for each k. Let k be the least integer such that

$$b_{i_1} \cdots b_{i_k} d \leq r < b_{i_1} \cdots b_{i_{k-1}} d.$$

Let $(j_1, \ldots, j_k) \in I^{\mathbb{N}}$ distinct from $(i_1, \ldots, i_k)$, so there is a least l such that $i_l \neq j_l$. Thus, $S_{i_l,\ldots,i_k} \subseteq S_{i_l}$ and $S_{j_l,\ldots,j_k} \subseteq S_{j_l}$ are separated by at least d. This implies that the sets $S_{i_1,\ldots,i_k}$ and $S_{j_1,\ldots,j_k}$ must be disjoint, separated by a distance of at least $b_{i_1} \cdots b_{l-1} d \geq b_{i_1} \cdots b_{k-1} d > r$. As $(j_1, \ldots, j_k)$ was arbitrary, any point within a distance r of x must be in $S_{i_1,\ldots,i_k}$, meaning $S \cap B(x, r) \subseteq S_{i_1,\ldots,i_k}$.

Let U satisfy $|U| < r$. If U intersects S, we can choose $x \in S$ such that $U \subseteq B(x, r)$. Therefore,

$$\nu(U) \leq \nu(S \cap B(x, r)) \leq \nu(S_{i_1,\ldots,i_k}) = (b_{i_1} \cdots b_{i_k})^s \leq d^{-s} r^s,$$

so by Theorem 2.8, we have $\mathcal{H}^s(S) > 0$ and $\dim_H S \geq s$. □

Combining these two theorems implies the most strict of our results for self-similar sets.

Theorem 4.13. *Let $\{f_1, \ldots, f_n\}$ be an IFS of similarities with attractor S and similarity value s. If $S_1, \ldots, S_n$ are pairwise disjoint, then $\dim_H S = s$ with $0 < \mathcal{H}^s(S) < \infty$.*

Proof. The value s in Theorems 4.11 and 4.12 corresponds to the similarity value for self-similar sets, so the result follows immediately. □

A less restrictive requirement that holds for all complete metric spaces is the strong open set condition. This permits some overlap between parts of the attractor, but requires a point where it coincides with a set that never does overlap. Therefore, the fractal is eventually almost disjoint with only an irregularity at a small scale, allowing us to apply Theorem 4.12. The irregularity disappears as we iterate the fractal infinitely.

Theorem 4.14. *Let $\{f_1, \ldots, f_n\}$ be an IFS of similarities with attractor S and similarity value s. If it satisfies the strong open set condition, so that there exists a non-empty bounded open set V such that*

$$\bigcup_{i=1}^{n} f_i(V) \subseteq V$$

with $V_1, \ldots, V_n$ pairwise disjoint and $S \cap V \neq \varnothing$, then $\dim_H S = s$.

Proof. We know from Theorem 4.11 that $\dim_H S \leq s$ with $\mathcal{H}^s(S) < \infty$. It remains to show the lower bound.

Let V be the set given by the strong open set condition. The intersection $S \cap V$ is non-empty, so there exists $x \in S \cap V$. The set V is open so there is $r > 0$ such that $B(x, r) \subseteq V$. For each $k \in \mathbb{N}$, $x \in S_{i_1,\ldots,i_k}$ for some $(i_1, \ldots, i_k)$ in I^k, and since $|S_{i_1,\ldots,i_k}| \to 0$ as $k \to \infty$, there exists l such that $|S_{i_1,\ldots,i_l}| < r$. As $x \in S_{i_1,\ldots,i_l}$, this implies $S_{i_1,\ldots,i_l} \subset B(x, r) \subseteq V$.

Applying f to either side, $S_{j,i_1,\ldots,i_l} \subset V_j$ for all j, and thus the $S_{j,i_1,\ldots,i_l}$ are pairwise disjoint. We can do this iteratively to get that $S_{j_1,\ldots,j_k,i_1,\ldots,i_l}$ are pairwise disjoint for all $k \in \mathbb{N}$.

The fractal corresponding to the similarities $f_{j_1} \circ \ldots \circ f_{j_k} \circ f_{i_1} \circ \ldots \circ f_{i_l}$ fulfils the assumptions of Theorem 4.12. Therefore, we have that $s_k \leq \dim_H S \leq s$, where s_k is the similarity value satisfying

$$1 = \sum_{I^k} (c_{j_1} \cdots c_{j_k} c_{i_1} \cdots c_{i_l})^{s_k} = (c_{i_1} \cdots c_{i_l})^{s_k} \sum_{I^k} (c_{j_1} \cdots c_{j_k})^{s_k}.$$

We know that

$$(c_{i_1} \cdots c_{i_l})^{-s_k} \leq (c_{i_1} \cdots c_{i_l})^{-\dim_H S} \leq (c_{i_1} \cdots c_{i_l})^{-s} < \infty.$$

But then

$$\begin{aligned}
(c_{i_1} \cdots c_{i_l})^{-s_k} &= \sum_{I^k} (c_{j_1} \cdots c_{j_k})^{s_k} \\
&\geq \sum_{I^k} (c_{j_1} \cdots c_{j_k})^{\dim_H S} \\
&= \sum_{I^k} (c_{j_1} \cdots c_{j_k})^{s} (c_{j_1} \cdots c_{j_k})^{\dim_H S - s} \\
&\geq \sum_{I^k} (c_{j_1} \cdots c_{j_k})^{s} (\max_i c_i)^{k(\dim_H S - s)} \\
&= (\max_i c_i)^{k(\dim_H S - s)}
\end{aligned}$$

must also be bounded as $k \to \infty$ and so $\dim_H S \geq s$. Hence, we have $\dim_H S = s$. □

Note that because we acquire the lower bound as a limit, it is no longer certain that the Hausdorff dimension is strict, as $\mathcal{H}^s(S)$ may be equal to 0.

4.3 Dimension in complete doubling spaces

The constraints on Theorem 4.14 can be loosened in certain common metric spaces, called doubling spaces, where the scaling of sets is controlled.

Definition 4.15. *A metric space (X, d) is a **doubling space** if there is a constant C such that for all $x \in X$ and $r > 0$, there exist $x_1, \ldots, x_C \in X$ that form a cover*

$$B(x, r) \subseteq \bigcup_{i=1}^{C} B\left(x_i, \frac{r}{2}\right).$$

Thus, any ball in a doubling space can be covered by a bounded number of balls with half the radius. The following lemma provides a similar bound for the number of smaller balls contained in any ball.

Lemma 4.16. *Let X be a doubling space with doubling constant C. For any $r > \rho > 0$, there exists a constant $c = c(C, \frac{r}{\rho})$ such that any ball of radius r contains at most c pairwise disjoint balls of radius ρ.*

Proof. Let $x \in X$ and $r > \rho > 0$. Consider the case where $\rho \geq \frac{r}{2}$. Let $\{B(x_i, \rho)\}_i$ be any collection of disjoint balls with $x_i \in B(x, r)$.

By applying the doubling property of X twice, we can find a cover

$$B(x, r) \subseteq \bigcup_{j=1}^{C^2} B\left(y_j, \frac{r}{4}\right).$$

Thus, for any index i there exists an index j such that $x_i \in B\left(y_j, \frac{r}{4}\right)$ and thus $B\left(y_j, \frac{r}{4}\right) \subseteq B(x_i, \rho)$ as $\rho \geq \frac{r}{2}$. Each $B\left(y_j, \frac{r}{4}\right)$ can only be contained in a single $B(x_i, \rho)$ as these are pairwise disjoint. Therefore, setting $c(C, \frac{r}{\rho}) = C^2$ if $\frac{r}{\rho} \leq 2$, we have that $i \leq C^2$.

Now consider $\frac{r}{2^n} \leq \rho < \frac{r}{2^{n-1}}$. By the same argument, any collection of disjoint balls $\{B(x_i, \rho)\}_i$ has $1 \leq i \leq C^{n+1}$. By induction, the statement is true for all $\frac{r}{\rho} > 1$. □

This will now enable us to provide a similar theorem for the number of intersecting balls. Note that because the constant is specifically dependent on the ratio between the radii, we are able to choose two multiples that cancel to form an independent constant.

Lemma 4.17. *Let X be a doubling space with doubling constant C and $\{U_i\}$ a family of disjoint open subsets. If every U_i contains a ball of radius $a_1 r > 0$ and is contained in a ball of radius $a_2 r > 0$, then there is a constant $c = c(C, a_1, a_2)$ such that any ball of radius r intersects at most c of the closures $\overline{U}_i$.*

Proof. Let $B(x, r)$ be any ball in X. As each U_i is contained in a ball of radius $a_2 r$, we know $|U_i| < 2a_2 r$. Hence, any $\overline{U}_i$ that intersects $B(x, r)$ is contained in $B(x, (1 + 2a_2)r)$. By Lemma 4.16, $B(x, (1 + 2a_2)r)$ contains at most

$$c = c\left(C, \frac{(1 + 2a_2)r}{a_1 r}\right) = c\left(C, \frac{1 + 2a_2}{a_1}\right) = c(C, a_1, a_2)$$

pairwise disjoint balls of radius $a_1 r$. Any $\overline{U}_i$ is contained in $B(x, (1 + a_2)r)$ only if the ball of radius $a_1 r$ it contains is at least contained in $B(x, (1+a_2)r)$. Therefore, $B(x, r)$ intersects at most c of the $\overline{U}_i$. □

We can now prove the final important result, which depends on the open set condition instead of the strong open set condition. A point of overlap between the open set and the fractal is no longer required, as instead of relying on disjointness, the doubling property of the space guarantees only a finite number of intersections. For these attractors, we can show that their Hausdorff dimension is strict.

Theorem 4.18. *Let $\{f_1, \ldots, f_n\}$ be an IFS of similarities on a doubling space with attractor S and similarity value s. If it satisfies the open set condition, so that there exists a non-empty bounded open set V such that*

$$\bigcup_{i=1}^{n} f_i(V) \subseteq V$$

with $V_1, \ldots, V_n$ pairwise disjoint, then $\dim_H S = s$ with $0 < \mathcal{H}^s(S) < \infty$.

Proof. As before, the upper bound is proven in Theorem 4.11. For the lower bound, define a mass distribution ν on S in the same way as the proof of Theorem 4.12 with $X = I$ and $p(i) = c_i^s$. We can show that it satisfies the condition of Theorem 2.8.

We first prove this for the case of a ball $B \subseteq X$ with radius $r < 1$. By the definition of the mass distribution, $\nu(B) = \nu(S \cap B)$. We will cover $S \cap B$ with sets $\overline{V}_{i_1,\ldots,i_k}$ from the open set condition to estimate its outer measure. In particular, we require that they satisfy the requirements of Lemma 4.17.

Thus, we will find finite sequences $(i_1, \ldots, i_k)$ with the following properties:

(i) each $V_{i_1,\ldots,i_k}$ contains a ball of radius $a_1 r$ and a ball of radius $a_2 r$;

(ii) the set of $V_{i_1,\ldots,i_k}$ is pairwise disjoint; and

(iii) $S \cap B \subseteq \bigcup_{i_1,\ldots,i_k} \overline{V}_{i_1,\ldots,i_k}$

Once we obtain an admissible cover, the measure will easily be estimated using Lemma 4.17 to satisfy the conditions of Theorem 2.8.

Let $(i_1, i_2, \ldots) \in I^{\mathbb{N}}$. The sequence $c_{i_1} \cdots c_{i_k} \to 0$ as $k \to \infty$, so there is a smallest k such that $c_{i_1} \cdots c_{i_l} < r$. As k is the smallest number for which this holds, $c_{i_1} \cdots c_{i_{k-1}} \geq r$, and so

$$(\min_i c_i) r \leq c_{i_k} r \leq c_{i_1} \cdots c_{i_k} \leq r.$$

Let J be the finite set of finite sequences $(i_1, \ldots, i_k)$ obtained by curtailing each $(i_1, i_2, \ldots)$ in this manner, where for each sequence $B \cap \overline{V}_{i_1,\ldots,i_k} \neq \varnothing$. This set satisfies the properties listed above:

(i) As the open set V is bounded and open, we can choose ρ_1 and ρ_2 such that V contains a ball of radius ρ_1 and is contained in a ball of radius ρ_2. For each $(i_1, \ldots, i_k) \in J$, the similarities $f_{i_1} \circ \ldots \circ f_{i_k}$ form a similarity of ratio $c_{i_1} \cdots c_{i_k}$, so each set $V_{i_1,\ldots,i_k}$ contains a ball of radius $c_{i_1} \cdots c_{i_k} \rho_1$ and is contained in a ball of radius $c_{i_1} \cdots c_{i_k} \rho_2$. We can set $a_1 r = (\min_i c_i)\rho_1 r \leq c_{i_1} \cdots c_{i_k} \rho_1$ and $a_2 r = \rho_2 r \geq c_{i_1} \cdots c_{i_k} \rho_2$.

(ii) For any pair of sequences in J, one sequence cannot contain the other completely, as the original sequences would have been curtailed at the same index. From the open set condition, $V_1, \ldots, V_n$ are pairwise disjoint, so $V_{i_1,\ldots,i_k,1}, \ldots, V_{i_1,\ldots,i_k,n}$ are also pairwise disjoint for all k and $(i_1, \ldots, i_k) \in I^k$. Applying this iteratively and using that $f(V) \subseteq V$, it follows that J is pairwise disjoint.

(iii) The closure of the open set $\overline{V}$ is closed and bounded. It follows directly from the doubling property of the space that $\overline{V}$ is totally bounded and therefore compact. Therefore, as $f(\overline{V}) \subseteq V$, Corollary 3.11 implies that $S = \bigcap_{k=0}^{\infty} f^k(\overline{V})$. This shows that $S \subseteq \overline{V}$, and so iteratively we have that $S_{i_1,\ldots,i_k} \subseteq \overline{V}_{i_1,\ldots,i_k}$ for each $(i_1, \ldots, i_k) \in I^k$ for any k. Furthermore, by construction any $x_{i_1,i_2,\ldots} \in S \cap B$ has a corresponding curtailed sequence $(i_1, \ldots, i_k) \in J$ such that

$$x_{i_1,\ldots,i_k,\ldots} \in \bigcap_{l=1}^{\infty} S_{i_1,\ldots,i_k,\ldots,i_l} \subseteq S_{i_1,\ldots,i_k} \subseteq V_{i_1,\ldots,i_k}.$$

It follows that

$$S \cap B \subseteq \bigcup_J S_{i_1,\ldots,i_k} \subseteq \bigcup_J \overline{V}_{i_1,\ldots,i_k}.$$

Therefore, we have a cover $\bigcup_J \overline{V}_{i_1,\ldots,i_k}$ of the ball B, which by Lemma 4.17 contains at most c sets. Hence,

$$\nu(B) \leq \nu\left(\bigcup_J \overline{V}_{i_1,\ldots,i_l}\right) = \mu^*(C_J),$$

as each $x_{i_1,i_2,\ldots} \in S$ is in $V_{i_1,\ldots,i_k}$ if the corresponding sequence $(i_1, i_2, \ldots)$ begins with $(i_1, \ldots, i_k) \in J$. We complete the estimation with

$$\nu(B) = \mu(C_J) = \sum_J (c_{i_1} \cdots c_{i_k})^s \leq \sum_J r^s \leq cr^s.$$

Since any set U with diameter $|U| < 1$ is contained in a ball with radius $r < 1$, it follows that $\nu(U) \leq c|U|^s$. By Theorem 2.8, we have the lower bound $\mathcal{H}^s(S) > 0$ and conclude that $\dim_H S = s$. □

4.4 Examples

We can use the theorems proved in this section to find a far easier solution to the dimension of the middle third Cantor set using a solely visual approach. In fact, it satisfies the conditions of all three results:

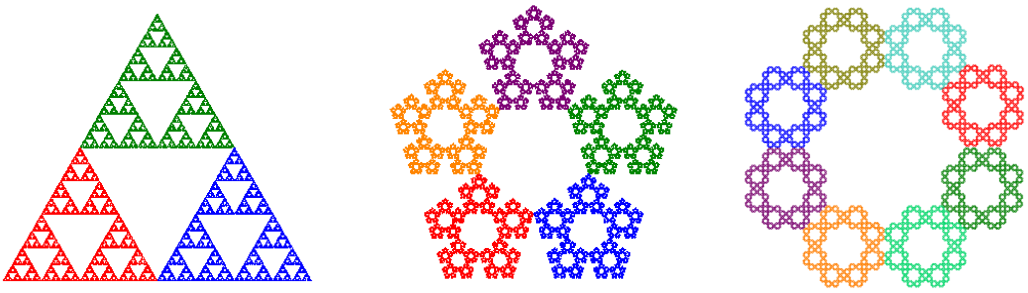
(i) for Theorem 4.13, we have that C_1 and C_2 are disjoint;

(ii) for Theorem 4.18, $\mathbb{R}$ is clearly a doubling space (we can cover any open interval with three open intervals of half length), and $(0, 1)$ satisfies the open set condition with

$$f_1((0,1)) \cup f_2((0,1)) = \left(0, \frac{1}{3}\right) \cup \left(\frac{2}{3}, 1\right) \subseteq (0,1);$$

(iii) for Theorem 4.14, we use the same open set with the additional condition that $C \cap (0,1) = C \setminus \{0,1\} \neq \varnothing$.

Any of these imply that the dimension is the solution of $\left(\frac{1}{3}\right)^s + \left(\frac{1}{3}\right)^s = 1$, confirming our previous result of $\log 2 / \log 3$. The first two also show that this dimension is strict, that is $0 < \mathcal{H}^s(C) < \infty$.



(a) Sierpinski triangle. (b) Sierpinski pentagon. (c) Sierpinski octagon.

Figure 2: Computer renders of Sierpinski n-gons for different n from [7].

Let us now consider a more interesting example in $\mathbb{R}^2$ detailed in [6]. The Sierpinski polygons seen in Figure 2 are a general case of the famous Sierpinski triangle. They are constructed by starting with a regular n-sided polygon with sides of length 1, which is then scaled so that n copies fit exactly inside the original, touching but not overlapping. This process is repeated recursively to obtain the fractals.

A Sierpinski n-gon can be represented by an IFS of similarities that uniformly scale and translate the points in the shape. To calculate the scaling ratio r_n, we consider one side of the starting polygon with length 1 and the two scaled copies lying along the edge in the next iteration, as seen in Figure 3. These two polygons touch at a single vertex or (if n is a multiple of 4) the entire edge. The sides of length r_n form the hypotenuses of a sequence of right triangles. By calculating the angles $\alpha_1, \alpha_2, \ldots$, we can find the lengths $d_1, d_2, \ldots$ of the bases.

As the interior angle of a regular n-sided polygon is $\beta = \frac{n-2}{n}\pi$, we can proceed inductively to find that

$$\begin{aligned}
\alpha_1 &= \pi - \beta = \frac{2\pi}{n} \\
\alpha_2 &= 2\pi - \beta - \frac{\pi}{2} - \left(\frac{\pi}{2} - \alpha_1\right) = \frac{2 \cdot 2\pi}{n} \\
&\ldots \\
\alpha_k &= \pi - \beta - \frac{\pi}{2} - \left(\frac{\pi}{2} - \alpha_{k-1}\right) = \frac{2k\pi}{n}.
\end{aligned}$$

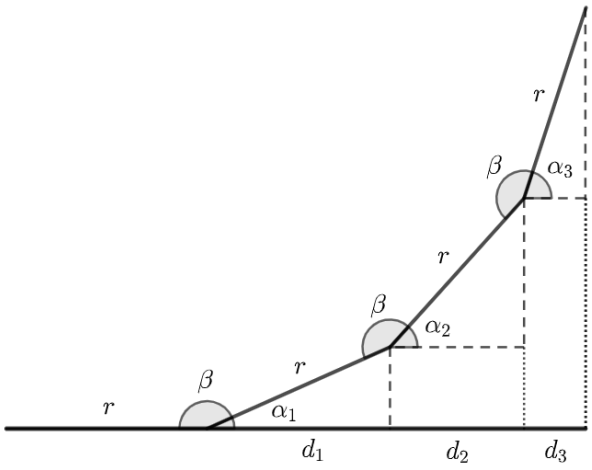


Figure 3: The calculation of the similarity ratio for an arbitrary n-gon.

Each of these angles is acute, so

$$\alpha_k = \frac{2k\pi}{n} \leq \frac{\pi}{2}$$

and thus $k \leq \frac{n}{4}$. For each of the angles α_k, the corresponding length $d_k = r_n \cos(\alpha_k)$, and similarly for the other symmetric polygon. Therefore, comparing the lengths to the original side length,

$$2\left(r_n + \sum_{k=1}^{\lfloor n/4 \rfloor} r_n \cos\left(\frac{2k\pi}{n}\right)\right) = 1$$

and hence

$$r_n = \frac{1}{2\left(1 + \sum_{k=1}^{\lfloor n/4 \rfloor} \cos\left(\frac{2k\pi}{n}\right)\right)}.$$

Regarding the translations, we can position the initial polygon with its centre at the origin and one vertex on the x-axis to obtain a clean formula. Each scaled polygon has a vertex that is mapped to a vertex of the original polygon. If $R_n = \frac{r_n}{2}\csc(\frac{\pi}{n})$ is the distance from the centre to a vertex, then the translation distance is thus $R_n(1 - r_n)$. Meanwhile, for $k = 1, \ldots, n$, the corresponding angles are $\frac{2k\pi}{n}$. Hence, the kth transformation of the IFS is

$$f_k(\mathbf{x}) = \begin{bmatrix} r_n & 0 \\ 0 & r_n \end{bmatrix} \mathbf{x} + R_n(1 - r_n) \begin{bmatrix} \cos\left(\frac{2k\pi}{n}\right) \\ \sin\left(\frac{2k\pi}{n}\right) \end{bmatrix}.$$

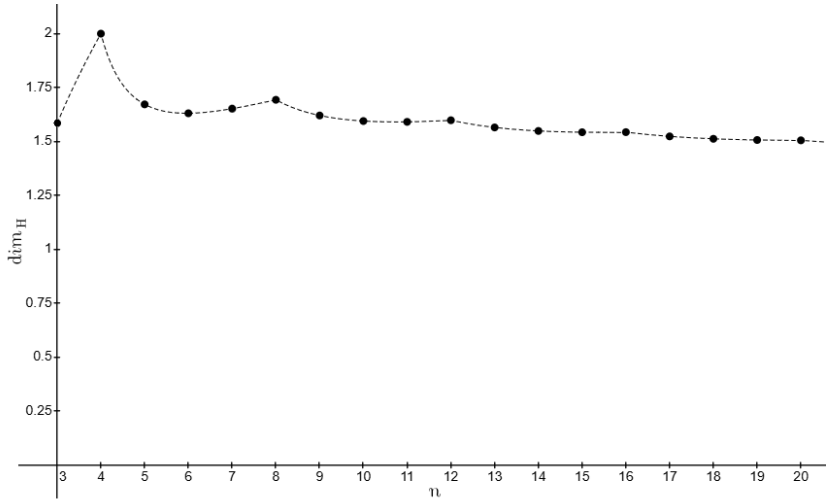


Figure 4: A graph of how the dimension of Sierpinski polygons changes with the number of sides.

Using this IFS, the Hausdorff dimension follows easily. Since the vertices of the shapes touch, the similarities no longer map to disjoint sets, so Theorem 4.13 no longer applies. However, the interior of the original polygon satisfies the open set condition, as the images are open and thus disjoint. With the additional facts that the open set intersects the fractal and that $\mathbb{R}^2$ is doubling, we can apply Theorems 4.14 and 4.18, respectively. As there are n similarities each with ratio r_n, the dimension of a Sierpinski n-gon is $-\log n/\log r_n$. The triangle, pentagon and octagon in Figure 2, for instance, have dimensions approximately equal to 1.58496, 1.67228, and 1.69343, respectively.

Figure 4 shows how the dimension changes with the number of sides. It gradually decreases as the area covered by the polygons decreases, forming spikes where polygons meet along an entire edge. This is due to the behaviour of r_n, which is a piecewise smooth, decreasing function whose derivative has discontinuities where the floor function jumps in value. Geometrically, the polygons fit together more efficiently when the angle between their edges decreases. It can be shown that the spikes decrease in magnitude and r_n is asymptotically equivalent to $\frac{\pi}{n}$. Therefore, the dimension

$$-\frac{\log n}{\log r_n} \sim \frac{\log n}{\log n - \log \pi} = \left(1 - \frac{\log \pi}{\log n}\right)^{-1} \sim 1 + \frac{\log \pi}{\log n} \to 1$$

as $n \to \infty$ with logarithmic convergence. The shape itself approaches a circle.

5 Conclusion

Fractals are a highly interesting class of mathematical objects with a number of surprising properties that have motivated the development of new analytical tools. This paper has explored foundational ideas in the study of fractal geometry, notably Hausdorff dimension and iterated function systems. These have been constructed with an emphasis on generality, extending the theory beyond typical applications in $\mathbb{R}^N$.

The highlights of the paper are the final two theorems on self-similar sets. These are essential results in the field, enabling the characterisation of many popular fractals in a simple and elegant manner. In the rigorous proof of these theorems, one can appreciate both the intersections with measure theory as well as the relationship between fractals and their underlying metric space.

There are several immediate areas for expansion. Other definitions of fractal dimension, such as box dimension and packing dimension, have analogous results relating to self-similar sets. There also exist further refinements of and extensions to the main theorems, such as the weak separation property introduced by Lau and Ngai [8]. Finally, the Hausdorff dimension of more complex fractals, such as self-affine sets, inhomogeneous attractors, and sets with graph self-similarity, may be explored using more advanced techniques.

References

[1] Kenneth Falconer. 2003. *Fractal Geometry: Mathematical Foundations and Applications* (2nd ed.). John Wiley & Sons Ltd, Chichester.

[2] Jesper van den Eijnden. 2018. *Fractal Dimension of Self-Similar Sets.* Radboud University. Retrieved 1 November, 2024 from `https://www.math.ru.nl/~mueger/THESES/Jesper_van_den_Eijnden_Bachelor_2018.pdf`

[3] Gerald Edgar. 2008. *Measure, Topology, and Fractal Geometry* (2nd ed.). Springer Science & Business Media, New York.

[4] Andreas Schief. 1996. Self-Similar Sets in Complete Metric Spaces. *Proc. Amer. Math. Soc.* 124, 2 (Feb. 1996), 481–490.

[5] Zoltán M. Balogh and Heiner Rohner. 2007. Self-Similar Sets in Doubling Spaces. *Illinois Journal of Mathematics* 51, 4 (Winter 2007), 1275–1297.

[6] Steven Schlicker and Kevin Dennis. 1995. Sierpinski n-gons. *Pi Mu Epsilon Journal* 10, 2 (1995), 81–89.

[7] Larry Riddle. 2022. *Classic Iterated Function Systems.* Agnes Scott College. Retrieved 8 February, 2024 from `https://larryriddle.agnesscott.org/ifs/ifs.htm`

[8] Ka-Sing Lau and Sze-Man Ngai. 1999. Multifractal Measures and a Weak Separation Condition. *Advances in Mathematics* 141, 1 (1999), 45–96.

[9] Mona Hazeem. 2022. *Similarities and Hausdorff Dimension in Fractal Geometry.* Stockholm University. Retrieved 1 November, 2024 from `https://kurser.math.su.se/pluginfile.php/16103/mod_folder/content/0/2022/2022_K1_report.pdf`

[10] Katie Barich. 2011. *Proving Completeness of the Hausdorff Induced Metric Space.* Whitman College. Retrieved 1 November, 2024 from `https://www.whitman.edu/documents/Academics/Mathematics/SeniorProject_KatieBarich.pdf`